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Thalavapalayam, Karur - 639 113, TAMILNADU.



# **Gesture Controlled Contactless Switch Using Camera Technology**

**Guided By**

**Mrs.M.Sentamilselvi, M.E.,(Ph.D).,**

**Presented By**

**MOHAMED SALMAN AL FARISH A(927623BEC135)**

**MOHAMED AATHIL J (927623BEC136)**

**MUKESH M (927623BEC141)**

# Introduction

With the rapid growth of smart homes and automation technologies, there is an increasing demand for contactless and intuitive methods to control electrical appliances. Traditional control systems rely on physical switches, remote controls or mobile applications, which require touch or external devices and may not always be convenient or hygienic. To address these limitations, this project presents a Gesture Controlled Contactless Switch using a Camera, which enables users to operate appliances using simple hand gestures. By combining computer vision (OpenCV) with an ESP32 microcontroller and relay module, the system detects finger-count gestures through a camera and wirelessly controls multiple devices. This approach provides a safe, low-cost and user-friendly solution suitable for smart homes, hospitals, laboratories and assistive applications.



## Objectives

1. Develop a touch-free smart home control system using hand gestures.

This reduces dependency on physical switches and enhances user convenience by enabling natural, intuitive and hygienic control of appliances.

2. Implement real-time gesture recognition using OpenCV and MediaPipe.

The system accurately detects finger-count gestures and converts them into control signals for the ESP32 microcontroller.

3. Ensure reliable activation of multiple home appliances through ESP32 and relay modules.

By mapping gestures to individual relay channels, the system enables safe, fast, and seamless switching of electrical loads.

## Scope of the Project

1. Provides hands-free control for lights, fans and other household appliances.

This enhances safety by minimizing physical contact and improves accessibility for elderly individuals and users with limited mobility.

2. Offers scalability to integrate additional gestures and more relay-controlled devices.

The system can be expanded for smart home, smart office and assistive technology applications with minimal hardware modifications.

3. Supports future enhancement through IoT connectivity and mobile-based monitoring.

Cloud control, automation scheduling and AI-based gesture improvement can be added to transform it into a complete smart home ecosystem.



# Literature Review

Recent research in smart home automation highlights the increasing use of computer-vision-based interaction systems to enhance convenience, accessibility, and safety. Several works demonstrate effective hand-gesture detection and IoT-based appliance control, but major limitations still exist in real-time reliability, system integration, and natural user interaction.

## **The Following Studies Summarize the Existing Contributions:**

- Vision-based gesture recognition systems focus mainly on finger detection and image processing, but many lack seamless integration with IoT hardware for real appliance control.
- ESP32 and microcontroller-based automation platforms have been widely used for Wi-Fi-enabled switching, yet most solutions rely on touch-based mobile apps or sensors rather than intuitive hand gestures.
- Computer-vision frameworks such as OpenCV and MediaPipe enable accurate landmark extraction, but existing works often do not address lighting variation, background noise, or consistent multi-gesture mapping for multi-device control.



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# Existing System

In the existing home automation systems, appliances are operated mainly through physical wall switches, remote controls or mobile applications connected via Wi-Fi or Bluetooth. These methods require direct touch or the use of external devices, which can be inconvenient, unhygienic and difficult for elderly or physically challenged users. Most current systems do not support natural, hands-free interaction and lack real-time gesture-based control. Furthermore, they offer limited automation and cannot operate appliances using intuitive human gestures, making them less efficient and not fully aligned with modern smart home needs.

# Problem Statement

Traditional home automation systems rely on physical switches or mobile apps, which require touch interaction and can be inconvenient, unhygienic or inaccessible for elderly and disabled users. These systems also lack a natural, intuitive way to control multiple appliances quickly. There is a need for a touch-free, efficient and reliable method to operate home devices without physical contact. Our project addresses this problem by using computer vision–based hand gesture recognition and an ESP32-controlled relay system to provide a smart, touchless solution for operating household appliances.

# Proposed System

The proposed system introduces a touch-free smart home automation solution that uses computer vision to recognize hand gestures and control electrical appliances. A webcam captures real-time images, which are processed using OpenCV and MediaPipe to detect the number of fingers shown by the user. These gesture commands are then transmitted to an ESP32 microcontroller, which activates the appropriate channel on a 4-channel relay to switch appliances ON or OFF. This system provides intuitive, hands-free control, enhances safety and accessibility and eliminates the need for physical switches or mobile applications, enabling a modern, efficient and user-friendly smart home experience.





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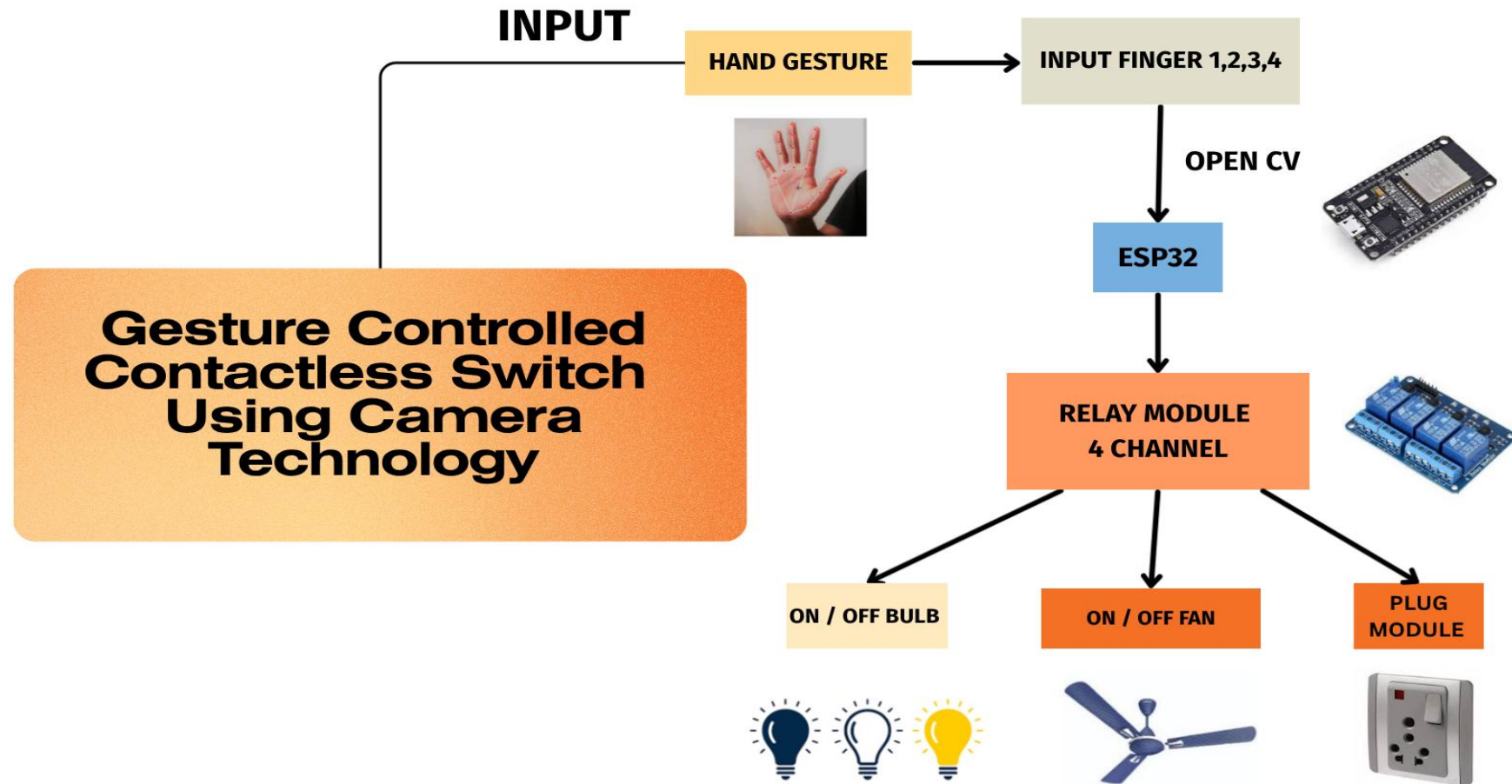
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## Block Diagram







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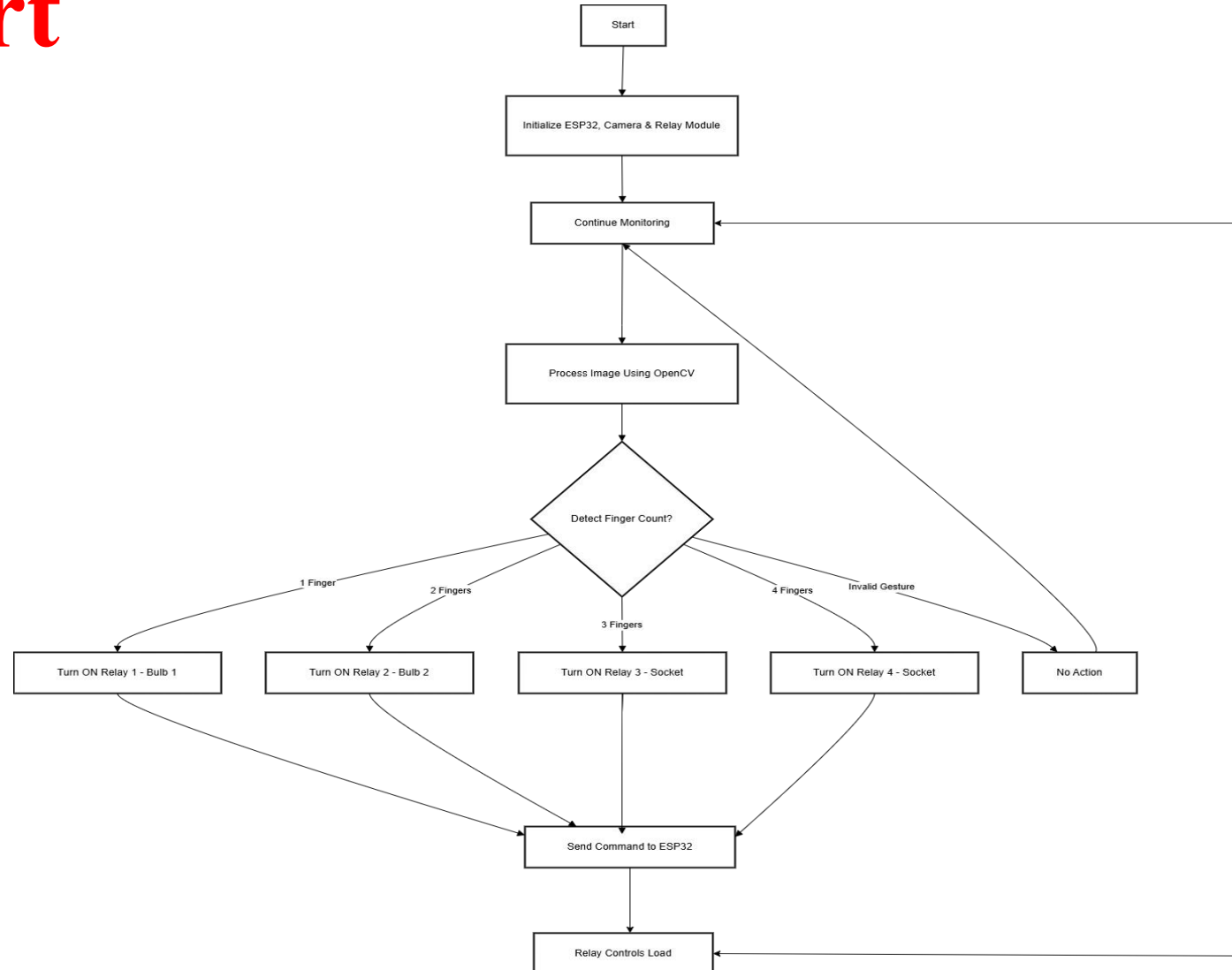
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# Flowchart





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# Methodology

- **Real-time image acquisition:** A webcam continuously captures live video frames of the user's hand and sends them to the computer for processing.
- **Hand landmark detection:** OpenCV and MediaPipe are used to detect the hand region and extract 21 key landmarks, enabling accurate identification of finger positions.
- **Gesture classification:** The system analyzes the detected landmarks to determine the number of fingers raised and classifies the gesture into a specific control command.
- **Wireless command transmission:** The recognized gesture is sent to the ESP32 microcontroller through serial or Wi-Fi communication, ensuring low-latency interaction.
- **Appliance control via relay module:** The ESP32 maps the received gesture to the corresponding relay channel and switches the connected electrical appliance ON/OFF.
- **System feedback and validation:** Visual or serial output confirms the executed action, ensuring reliable and user-friendly operation of the gesture-controlled automation system.



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## Hardware

- **ESP32 Microcontroller** – Acts as the central controller that receives gesture commands, processes input signals, and activates the relay module for appliance control.
- **Webcam / USB Camera** – Captures real-time video frames of the user's hand for gesture recognition through OpenCV and MediaPipe.
- **4-Channel Relay Module** – Controls the ON/OFF switching of home appliances based on commands sent by the ESP32.
- **Bulb / Fan / Load Devices** – Electrical appliances used to demonstrate the gesture-based switching functionality.
- **Connecting Wires & Breadboard** – Used for system wiring, routing signals, and assembling the prototype safely and efficiently.

## Software Used

- **Python (OpenCV & MediaPipe)** – Used for image processing, hand landmark extraction, and gesture classification in real time.
- **Arduino IDE** – Used for coding, compiling, and uploading the ESP32 firmware that maps gesture commands to relay operations.
- **OpenCV Library** – Performs video capture, frame processing, contour detection, and landmark segmentation.



# Simulation Results

sketch.ino • diagram.json • libraries.txt • Library Manager

```
1 // ESP32_WebSocket_Relays.ino
2 #include <WiFi.h>
3 #include <WebSocketsServer.h>
4
5 const char* ssid = "vivo V27";
6 const char* password = "salman786";
7
8 WebSocketsServer webSocket = WebSocketsServer(81); // WS port 81
9
10 // Use GPIO 14, 27, 26, 25 on ESP32
11 int relayPins[] = {14, 27, 26, 25}; // change if you want other pins
12 const int RELAY_COUNT = 4;
13
14 void handleMessage(uint8_t num, char * payload) {
15     // payload expected as 4-char string: "IMRP" each char '1' or '0'
16     Serial.print("WS msg from ");
17     Serial.print(num);
18     Serial.print(": ");
19     Serial.println(payload);
20
21     String s = String(payload);
22     if (s.length() >= RELAY_COUNT) {
23         for (int i = 0; i < RELAY_COUNT; ++i) {
24             char c = s.charAt(i);
25             if (c == '1') {
```

Simulation

▶ + ⋮





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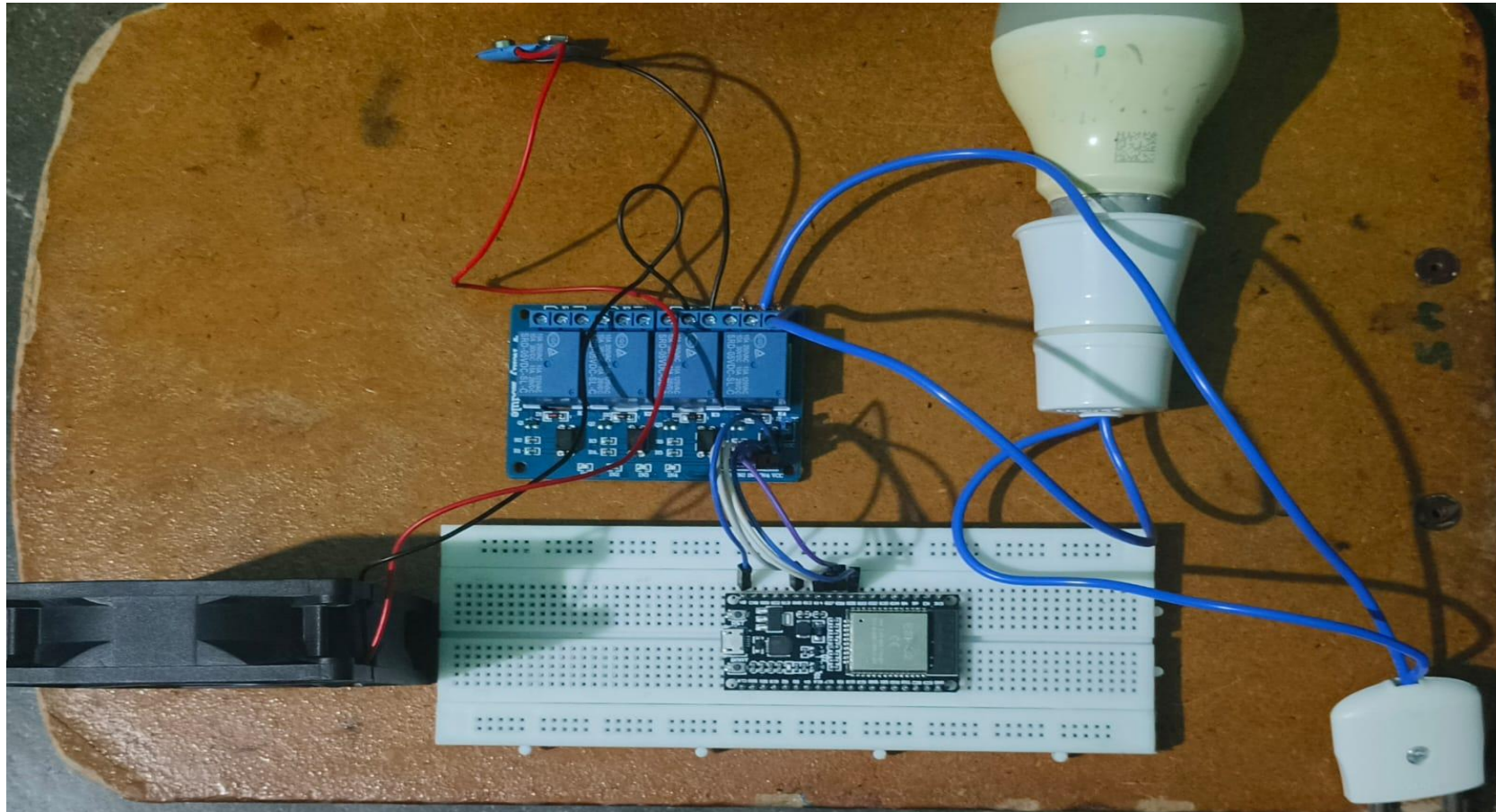
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# Hardware Prototype





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# Results & Discussion

The developed gesture-based smart home automation system successfully recognized hand gestures in real time using OpenCV and MediaPipe and accurately controlled electrical appliances through the ESP32 and relay module. Testing under different lighting and background conditions showed consistent gesture detection with minimal delay, demonstrating the reliability of the vision-based approach. The ESP32 responded instantly to each gesture, ensuring smooth ON/OFF switching of loads such as bulbs and fans. Overall, the results confirm that the proposed system provides a fast, touchless and user-friendly method for home automation, improving accessibility and safety compared to traditional switch-based systems.





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# Applications

## **Smart Homes**

- Enables touch-free control of lights, fans, and household appliances, improving convenience and enhancing accessibility for elderly and physically challenged users.

## **Healthcare & Assisted Living**

- Provides a hygienic, contactless control method that reduces infection risk and supports patients who have limited mobility or cannot operate physical switches.

## **Smart Offices & Workspace**

- Allows gesture-based operation of electrical devices such as projectors, lighting systems, and presentation equipment, creating a modern and efficient interaction environment.

## **Home Automation Research & Education**

- Serves as an effective experimental platform for learning IoT, computer vision, and embedded systems, supporting academic projects and prototype development.



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# Advantages

- Touch-Free Operation
- Real-Time Gesture Recognition
- Improved Accessibility

# Limitations

- Lighting Dependency
- Camera Position Sensitivity
- Processing Requirement

# Project Outcome

## PROJECT OUTCOME : CONFERENCE PAPER

**ICISS 2026**

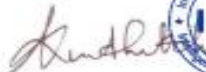
**8<sup>th</sup> International Conference on  
Intelligent Sustainable Systems  
(ICISS 2026)**  
4-6, March 2026 | [icoiss.com/2026/](https://icoiss.com/2026/) | [icoiss2k20@gmail.com](mailto:icoiss2k20@gmail.com)

**Acceptance Letter**

**Title: Gesture Controlled Contactless Switch using Camera**  
**To: Mrs.M.Sentamilselvi,Mohamed Salman Al Farish A,Mohammed Aathil J,  
Mukesh M**  
**Paper ID: ICISS – 097**

With the heartiest congratulations, we are happy to inform you that based on double blind review process and the recommendations of the conference review committee, your paper mentioned above has been accepted for publication and presentation at the International Conference on Intelligent Sustainable Systems [ICISS 2026].

ICISS 2026 gives due recognition to great achievements of students, researchers and industrialists in the promotion and effective utilization of their research works. For more information on the conference kindly visit ICISS 2026 Conference Website:  
<https://icoiss.com/2026/>



**Dr. R. Karthik**  
Ganesh Organizing Secretary.



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# Future Scope

The proposed gesture-based home automation system can be further enhanced by integrating IoT connectivity for cloud-based monitoring and remote control. Advanced AI models or deep-learning techniques can be incorporated to improve gesture accuracy under varying lighting and background conditions. The system can also be expanded to support more complex gestures, voice integration and multi-room automation. Additionally, incorporating low-power hardware and edge AI processing can make the solution more efficient and suitable for commercial smart home applications, healthcare environments, and assistive living systems.



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# Conclusion

The gesture-based smart home automation system successfully demonstrates an efficient, intuitive, and touch-free method for controlling household appliances using OpenCV, MediaPipe and an ESP32 microcontroller. The system accurately recognizes hand gestures in real time and reliably activates the corresponding relay channels, offering a safer and more accessible alternative to traditional switch-based operation. Testing under various conditions confirms the robustness and responsiveness of the proposed method. Overall, this work highlights the potential of computer vision and IoT integration in advancing modern smart home technologies and creating user-friendly, hygienic and intelligent automation solutions.





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